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11 October 1974
SUPERSEDING
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MILITARY SPECIFICATION

PYROMETERS, OPTICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of optical pyrometers.

1.2 Classification. The pyrometers shall be of the following types, as specified (see 6.2):

Type I - Absorption.
Type II - Disappearing filament.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-L-164	- Leather Dressing, Mildew-Preventive.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Military

MIL-B-18	- Batteries, Dry.
MIL-P-116	- Preservation-Packaging, Methods of.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

FSC 6685

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(Copies of specifications and standards required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The pyrometers shall consist essentially of a telescope and incandescent lamp, battery, and temperature scales.

3.2 First article (preproduction model). The supplier shall furnish one or more pyrometers as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce pyrometers that comply with this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

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3.4 Telescope. The telescope shall include an ocular lens, objective lens, eyeshield, electric switch, lamp, and temperature scales.

3.4.1 Ocular lens. The ocular lens shall have good optical correction, and shall be capable of being focused so that either a test mark in the field of vision or the lamp filament appears clear and sharp when the pyrometer is adjusted for maximum light intensity.

3.4.2 Objective lens. The objective lens shall be adjustable for observation of objects at various distances.

3.4.3 Eyeshield. A soft, flexible eyeshield, conforming to the face and large enough to be used with spectacles, shall be provided on the telescope.

3.4.4 Switch. A normally open switch, operated by pressure of the index finger or when the telescope is gripped, shall be located on the housing of the telescope. When the switch is closed the lamp shall light.

3.4.5 Lamp. The lamp shall be an incandescent filament type and shall retain calibration for not less than 300 hours of normal use.

3.5 Temperature scales. Temperature scales shall be graduated in degrees Fahrenheit (°F) and shall cover an overall temperature range of 1400° F to not less than 3200° F. Each scale shall have an accuracy of plus or minus 0.50 percent of the temperature measured.

3.6 Rheostat. A rheostat shall be provided to control the brillance of the lamp.

3.7 Batteries. The pyrometer shall operate on dry batteries or rechargeable nickel-cadmium batteries as specified (see 6.2).

3.7.1 Dry batteries. Dry batteries shall conform to MIL-B-18, type optional. Dry batteries shall not be furnished with the pyrometer.

3.7.2 Nickel-cadmium batteries. When nickel-cadmium batteries are furnished with the pyrometer, the batteries shall be provided with a charger which will recharge the batteries from any 120 V, 60 Hz current.

3.8 Carrying strap. Each pyrometer shall be provided with a nylon or leather carrying strap. The strap shall be provided with a snap at each end for fastening to the pyrometer. The length of the strap shall permit wearing the pyrometer around the neck.

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3.9 Carrying case. Each pyrometer shall be furnished with a sturdy carrying case of plastic, hardwood, or leather. The case shall be provided with a nylon or leather strap to permit hanging the case from the operator's neck. One strap for the pyrometer and the case, or separate straps for the case and pyrometer, may be provided. The interior of the case shall be blocked and cushioned to prevent movement of the instrument in the case and to prevent accidentally turning on the switch (see 3.4.4).

3.9.1 Leather. Any leather components, if used, shall be treated with a mildew-resistant dressing conforming to O-L-164, type optional.

3.9.2 Plastic. Plastic, if used, shall be of a high-impact type entirely suitable for use in foundries.

3.10 Instructions. Complete instructions for operation and maintenance of the pyrometer shall be provided when specified (see 6.2) and shall be placed in the carrying case.

3.11 Finishing. Each pyrometer and its case shall be finished in accordance with the supplier's commercial practice.

3.12 Identification marking. Each case and each pyrometer shall be identified in accordance with MIL-STD-130.

3.13 Type I, Absorption. Type I pyrometers shall be as shown on Figure 1, and shall include a calibrated, circular optical absorption wedge; a totally reflecting prism; and, an ammeter. Two temperature scales, one for the lower range and one for the higher range, shall indicate the true temperature of the radiating solids; and a third scale shall indicate the true temperature of molten iron and steel. The pyrometer shall have a focal length ranging from 3 feet or less to infinity. A master lamp, marked to permit identification, shall be provided for calibrating the service lamp.

3.13.1 Optical absorption wedge. The circular optical absorption wedge, placed between the objective lens and the ocular lens, shall vary the apparent brightness of the observed hot body to match the brightness of the lamp.

3.13.2 Prism. The prism shall transmit the light from the absorption wedge to the ocular lens for comparison with the light from the lamp.

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3.13.3 Ammeter. The ammeter, graduated at intervals of 1 milliampere, shall be mounted within the telescope and shall be connected into the lamp circuit. A correction screw shall be provided for adjusting the ammeter pointer to the zero position.

3.14 Type II, Disappearing filament. Type II pyrometers shall be as shown on Figure 2 and shall consist of the telescope and a temperature-indicating potentiometer. Absorption screens shall be provided for the high-temperature range to decrease the apparent brightness of the object. The temperature scales, one for the lower range and one for the higher range, shall indicate the temperature of radiating solids; and, in addition, compensating screens shall be provided for indicating the true temperature of molten iron and steel. The pyrometer shall have a focal length ranging from 7 inches to infinity. Leather or nylon carrying straps and a telescope holder shall be attached to the potentiometer.

3.14.1 Potentiometer. The temperature-indicating potentiometer shall vary the brightness of the incandescent lamp filament; balance the lamp voltage against the electromotive force of a standard battery or group of batteries; and indicate directly on the temperature scales the temperature of the hot body. A slide wire for balancing the potentiometer and a null point galvanometer as a balance indicator shall be provided. A small knob shall control the slide wire. A large knob shall control a rheostat directly and in addition shall control the slide wire through a friction clutch and shall compensate for the effect on the balance point of adjusting the rheostat.

3.15 Workmanship. The pyrometers and the cases shall be free from such defects as cracks, scratches, or other defects which could impair their operation or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction pyrometers shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction pyrometers shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level III.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Acceptable quality level (AQL) shall be 1.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. Pyrometers shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

Major

101. Eyeshield not as specified.
102. Switch not as specified.
103. Carrying strap not as specified.
104. Carrying case not as specified.
105. Instructions missing or not complete when required.
106. Identification marking not as specified, illegible, or missing.
107. Workmanship not as specified.

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4.5.2 Tests. Samples selected in accordance with 4.4.1 shall be subjected to the following test. Failure of the test shall be cause for rejection.

4.5.2.1 Accuracy. The test shall consist of reading the temperature of a pyrometer strip lamp at temperatures of 1400° F, 1800° F, 2500° F, and 3200° F. The readings taken with the sample pyrometers shall be compared with the readings taken with an optical pyrometer certified by the National Bureau of Standards. The readings of both pyrometers shall be taken simultaneously and at the same spot on the lamp filament. Readings not within the tolerances specified in 3.5 shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Methods; materials, and containers not as specified. Each incorrect method, material, or container shall constitute one defect.
109.	Pyrometer and instructions not placed in carrying case as specified for Level A or C.
110.	Closure and strapping of the boxes not as specified for Level A or C.
111.	Marking illegible, incorrect, or incomplete.

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5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified to prove prior to starting production packaging and packing that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements of this specification. Examination shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be prepared utilizing either the preproduction model pyrometer or a production pyrometer. If the preproduction model pyrometer is utilized, any preservative, packaging, and packing shall be removed by the supplier at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model pyrometer and production pyrometer.

5.2 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.2.1 Level A. Each complete pyrometer with instructions when furnished and contained in the case provided shall be preserved in accordance with MIL-P-116, Method IIb. The carrying case shall be the inner container and a fiberboard box conforming to PPP-B-636, W5c, shall be the outer container.

5.2.2 Level C. Each complete pyrometer with instructions when furnished shall be placed in the carrying case provided. The carrying case containing the items shall be packaged to afford protection against deterioration and damage during shipment from the supplier to the initial destination.

5.3 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.3.1 Level A. Pyrometers of like description, preserved and packaged as specified in 5.2, shall be packed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, Grade B, or PPP-B-621, Class 2. Closure and strapping of the boxes shall be in accordance with the appendix to the applicable box specification.

5.3.2 Level B. Pyrometers of like description, preserved and packaged as specified in 5.2, shall be packed as specified in 5.3.1 for Level A except the boxes shall be Domestic Type; or the boxes shall be of fiberboard construction conforming to PPP-B-636, V3c or equivalent. Fiberboard boxes shall be closed Method V of the box specification appendix.

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5.3.3 Level C. The pyrometers, preserved and packaged as specified in 5.2, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pyrometers are intended for use in determining the temperature of heated objects. Type I pyrometers are intended for use where portability is required. Type II pyrometers are intended for use where portability is a secondary consideration.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of pyrometer required (see 1.2).
(c) The time frame required for submission of preproduction model and number of models required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(e) Whether dry batteries or nickel-cadmium batteries are required (see 3.7).
(f) When complete instructions are required (see 3.10).
(g) Level of preservation and packaging and level of packing required (see 5.2 and 5.3).

6.3 Preproduction model. Any changes or deviations of production pyrometers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish pyrometers conforming to this specification.

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6.4 Preproduction pack. Any changes or deviations of production pa from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the pyrometers in accordance with this specification.

6.5 Information figures. Figures 1 and 2 show pyrometers which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other pyrometers which conform to this specification.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each pyrometer.

Custodian:

Army - ME

Preparing activity:

Army - ME

User activity:

Army - GL

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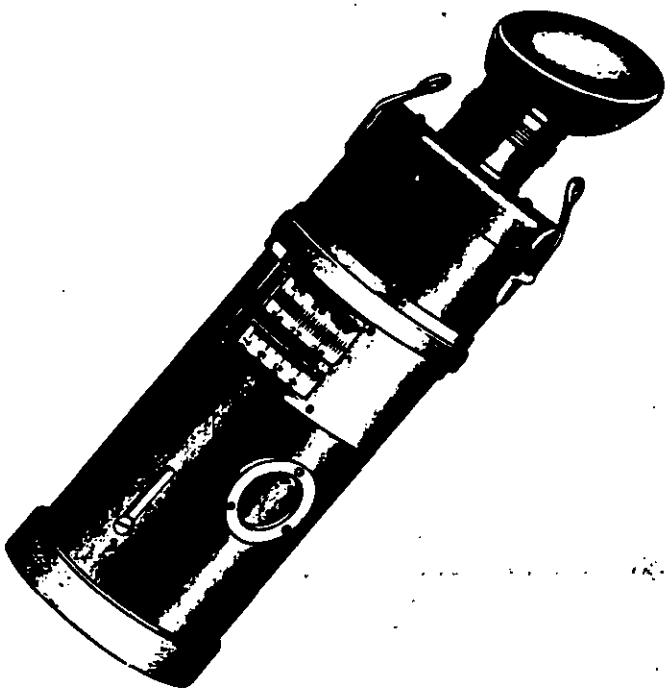


FIGURE 1. ABSORPTION WEDGE PYROMETER

X-2790

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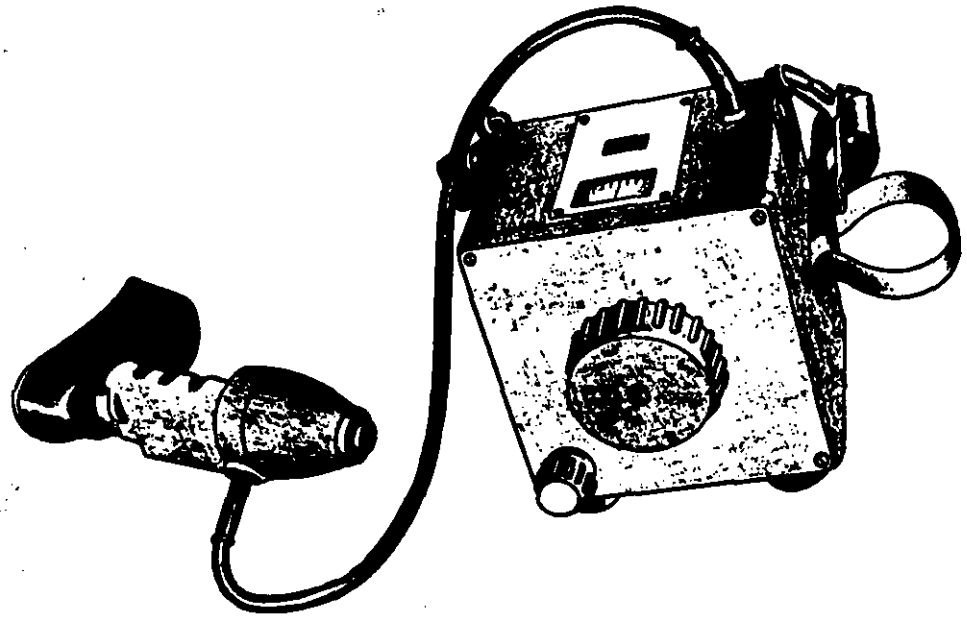


FIGURE 2. DISAPPEARING FILAMENT PYROMETER